The Last Straw: Benchmarking Computer-Use Agents Against Human Patience

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Abstract

Computer-use agents (CUAs) are evaluated on whether they eventually reach the correct end state, as judged by rubrics or deterministic state checks. Users judge them differently: an agent that needs three minutes and forty actions to change a setting that a person changes in five seconds is correct, but nobody would wait for it. We introduce LastStraw, a benchmark that measures when CUAs frustrate the people they act for. It contains 75 human-written tasks that people complete in seconds but that are prone to agent failure, such as dragging an item, choosing from a hover menu, picking a date or dismissing a dialog. For every task, human annotators record the effort it takes them and the tolerance they would extend to an agent: how long, and through how many unproductive actions, they would watch it before stopping it and doing the task themselves. A run succeeds only if it is correct and stays within this tolerance, so failing, looping, stalling and overrunning the human budget all count as frustration. We also define eight frustration patterns, with detectors, that explain why a run frustrates. Across M frontier models, the best model is correct on XX% of runs but succeeds within human tolerance on only XX%; correct runs take a median X.X× the human time, and looping appears in XX% of frustrated runs. Ranking models by tolerated success instead of correctness reorders N of the M models. The gap between agents and people is therefore larger than pass rates suggest. We release the tasks, human labels and evaluation harness.

1 Introduction

Computer-use agents (CUAs) operate graphical interfaces the way people do: they read the screen and act through the mouse and keyboard. In two years they have moved from research previews [1] to products [2, 3] and open models [4–6]. Their progress is tracked by benchmarks such as OSWorld [7], on which the best agent completed 12.24% of tasks at release against 72.36% for people. In a standardized evaluation, frontier agents now score up to 91.6% on the same benchmark [8], well above that human baseline.

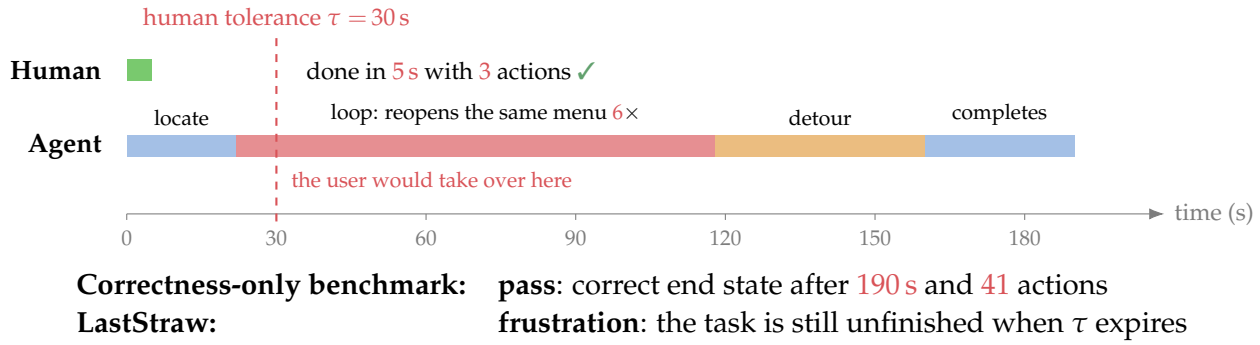


Figure 1. The same agent run under two evaluation regimes (illustrative, replace with a real trajectory). The task takes a person 5 seconds. The agent reaches the correct state, so a pass-rate benchmark counts a success, but it spends most of the run in a loop and finishes long after the point at which annotators say they would stop the agent and do the task themselves. LastStraw counts this run as a failure.

These benchmarks ask one question of each run: is the final state correct? A deterministic checker or a rubric compares the end state with the goal [7, 9–11], and the agent may use a fixed budget of steps (15 to 100 on OSWorld) with no limit on time. Under this protocol, a correct run counts the same whether it takes five seconds or five minutes. Measurements of agent efficiency show what this hides. On OSWorld, the best agents take 2.7–4.3× more steps than a person needs, with end-to-end latencies of tens of minutes on tasks that take people a few minutes [12]; on short UI-component tasks, even the fastest agent configuration takes 3.7× as long as a person [13].

A user experiences such a run differently. A person who hands a task to an agent compares the agent with doing the task themselves, and stops the agent once it takes longer, or flails more, than they are willing to watch. HCI research has long studied this experience as frustration. In diary studies, users lost between a third and a half of their time at the computer to frustrating episodes [14, 15]; delays beyond about ten seconds break a user's attention on the task [16, 17]; and people stop relying on an automated aid once they have seen it err [18, 19]. Studies of today's agents report the same experience: latency is the main complaint and a reason users take over control [20], and deployed agents loop and finish more slowly than the user would by hand [21]. For a user, a run they would have abandoned is a failed run, even if the agent would eventually have reached the goal.

The tasks most likely to exhaust a user's patience are often the ones people find easiest. Dragging a file, choosing from a hover menu, picking a date in a custom widget or closing a pop-up takes a person seconds, but it requires precise grounding, timing or continuous control, which agents still handle poorly [13, 22, 23]. This is Moravec's paradox [24] in the graphical interface. Most benchmarks emphasize long workflows across applications [7, 25, 26], in which such interactions are a small part of each task and their cost disappears into a single pass/fail verdict. Benchmarks that measure efficiency compare agents with the steps or time a person needs [12, 13, 27], but none asks how much more than their own effort people would accept from an agent before giving up on it.

We introduce LastStraw, a benchmark that measures frustration rather than correctness alone (Figure 1). It is named for the point at which a user's patience runs out and they take the task back. It contains 75 human-written tasks that people complete quickly (median XX s) and that centre on interactions prone to agent failure, in eight categories from continuous manipulation to interrupting dialogs (Table 1). For each task, n annotators record two labels. *Human effort* is the time and number of actions the task takes them. *Human tolerance* is how long, and through how many unproductive actions, they would watch

an agent before stopping it and doing the task themselves. A run succeeds only if the checker accepts the end state *and* the agent stays within both tolerance budgets. Everything else counts as frustration: an outright failure, a loop, a run stalled on deliberation, and a correct run that finishes after the user would have taken over. A taxonomy of eight frustration patterns, with automatic detectors, explains why each frustrated run failed (Table 2).

We evaluate M frontier models (§4). Three findings, to be confirmed against results. First, correctness overstates usability: the best model is correct on XX% of runs but succeeds within human tolerance on only XX%, while human annotators reach XX%. Second, agents remain far from human effort even when they succeed, taking a median X.X× the human time on correct runs, and ranking models by tolerated success instead of correctness reorders N of the M models. Third, frustration concentrates in a few patterns: looping and primitive failures account for XX% of frustrated runs, and higher reasoning effort raises correctness but lowers tolerated success.

Our contributions are:

- **LastStraw**, 75 human-written computer-use tasks that are easy for people, each with a deterministic checker and with human effort and tolerance labels from n annotators (§3.2–3.3).
- **A tolerance-aware evaluation protocol** that turns human patience into a pass/fail criterion, with the tolerated success rate, the frustration rate and the tolerance gap between correctness and tolerated success (§3.4).
- **A taxonomy of frustration patterns** with automatic detectors validated against human labels (§3.5).
- **An evaluation of M frontier CUAs** showing that headline finding (§4).

2 Related Work

Benchmarks for computer-use agents. Interactive benchmarks evaluate CUAs on desktop operating systems [7, 28–30], websites [9, 11, 31, 32], enterprise software [25, 33, 34] and mobile apps [10], alongside offline trajectory datasets [35] and grounding tests [22]. They judge the final state with deterministic checkers or rubric-based LLM judges [11, 36], and they differ mainly in domain and horizon, up to tasks that take a person more than an hour [26]. Time enters only as a step cap, and the cap itself shifts scores [4, 37, 38]; AndroidWorld derives its per-task cap from human step counts [10]. Robustness benchmarks add pop-ups and other everyday disruptions [23, 39], and PersonaCUABench finds that many fully completed runs still fall short of the requesting persona's satisfaction [40]. LastStraw keeps deterministic end-state checks, but a correct end state is not enough: the run must also stay within a human-labelled tolerance.

Efficiency, speed and reliability. A growing body of work argues that accuracy alone misrepresents agents. Accounting for cost [41, 42], consistency across repeated runs [43, 44] and benchmark rigour [45] each changes the conclusions drawn from pass rates. For CUAs, OSWorld-Human annotates a human trajectory for every OSWorld task and finds that the best agents take 2.7–4.3× more steps than necessary, with end-to-end latencies of tens of minutes [12]. cua-speedrun standardizes the infrastructure for timing CUAs and finds that frontier agents spend 86–127 s per OSWorld task [8]. ComponentBench pairs 2,910 programmatically generated UI-component tasks with human reference trajectories, and even its fastest agent configuration takes 3.7× as long as the matched human [13]. METR measures agent capability by the human completion time of the tasks agents can do [46], and ARC-AGI-3 scores

agents by their action efficiency relative to humans on abstract games [27]. These works use human effort as a reference for efficiency or difficulty. We also ask annotators how much more than their own effort they would tolerate from an agent, and we use that tolerance as a pass/fail boundary on realistic, human-written tasks.

Easy for people, hard for agents. Benchmarks made of tasks that people solve easily expose gaps that hard tasks hide, including web games [47], web browsing questions [48], abstract reasoning puzzles [27, 49, 50] and commonsense questions [51]. The pattern echoes Moravec's paradox [24]: the perception and motor skills that people exercise without effort are among the hardest to automate. LastStraw applies this design to ordinary interactions in ordinary applications, and asks both whether agents succeed at them and whether they do so before a person would give up.

Failure modes of agents. Taxonomies of agent failures [52, 53] and studies of overthinking, in which reasoning models spend far more computation on easy problems than they need [54, 55], explain why agents fail or waste effort. For GUI agents, repetitive loops and premature stopping recur as dominant failure modes [31, 56, 57], and AgentRewardBench labels repetitiveness alongside success and side effects [36]. These taxonomies are organized around the agent's internal errors. Our frustration patterns (§3.5) are organized around what the user sees, and they include patterns such as stalling and detours that a correctness criterion does not penalize at all.

User frustration and tolerance. HCI research defines frustration as the appraisal of an interface as an obstacle to a goal [58], and diary studies found that it costs users a third to a half of their time at the computer [14, 15]. Waiting is a central cause. Classic response-time limits place the end of uninterrupted attention at about ten seconds [16, 17, 59], web users tolerate about two seconds when retrieving information [60], and task performance and the intention to return drop with delays of a few seconds [61]. Expectations come from how long a task usually takes, so a result that takes far longer than expected frustrates [62], and direct-manipulation tasks require faster responses than conversational ones [63]. Human effort can be predicted from operator times with the Keystroke-Level Model and GOMS [64, 65]; we measure it directly instead. People stop relying on automation after they observe it err [18, 19, 66], and guidelines for human-AI interaction call for efficient dismissal and correction of an assistant [67, 68]. Recent studies of LLM agents find that latency is a main complaint and a reason users take over [20, 69], that deployed agents loop and run slower than doing the task by hand [21, 70], and that oversight depends on when control passes between user and agent [71–73]. Tolerance for delay also depends on the task [74]. We therefore label tolerance per task instead of assuming a global limit, and we use it as the boundary between an acceptable and a frustrating run.

3 LastStraw

LastStraw pairs each task with two kinds of human labels: how much effort the task takes a person, and how much time and how many actions a person would give an agent before stopping it and doing the task themselves (Figure 2). We describe the design principles (§3.1), the tasks (§3.2), the human labels (§3.3), the metrics (§3.4) and the frustration patterns (§3.5).

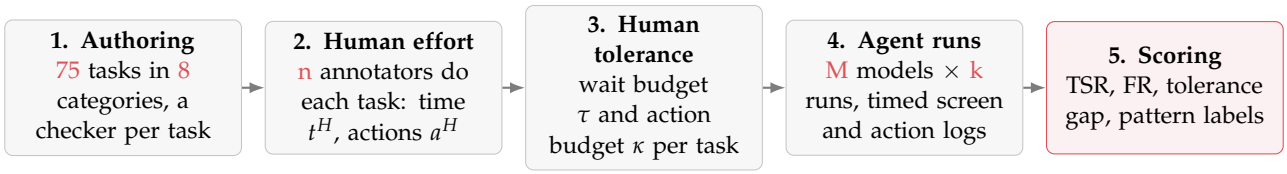


Figure 2. Construction and evaluation pipeline of LastStraw. Human labels (stages 2–3) set a per-task budget, and a run passes only if it reaches the correct end state within that budget (§3.4). [TODO: Replace with the final pipeline figure.]

3.1 Design principles

Easy for people. Every task is one that a person familiar with the application completes quickly and reliably from the task text alone. The median human completion time is XX seconds, and XX% of tasks take under XX seconds (§3.3).

Frustration-prone for agents. Each task centres on an interaction that is routine for people but where agents plausibly stall, loop or take a detour, such as a drag, a hover menu, a custom widget or an interrupting dialog (Table 1).

Correctness is necessary but not sufficient. Every task has a deterministic checker for the end state, as in existing benchmarks [7, 9]. A correct run that exceeds the task's human budget still fails.

The user's view of time. Budgets are stated in wall-clock time and in actions, because a user waiting for an agent perceives both how long it takes and how much it flails. [TODO: Decide whether model latency counts fully toward wall-clock time or whether we also report a latency-normalized variant; see §6.]

3.2 Tasks

Authoring. N annotators wrote the 75 tasks following written guidelines. A task must (i) have a single unambiguous goal stated in one or two sentences, (ii) take a person under XX seconds, (iii) have an end state that a script can verify, and (iv) avoid logins, payments, CAPTCHAs and actions on live third-party services. [TODO: Describe the authoring guidelines and review rounds in Appendix A.]

Categories. We group tasks by the interaction that makes them frustration-prone (Table 1): continuous manipulation, scrolling and hidden content, transient UI, custom widgets, text selection and editing, interruptions, keyboard and window management, and visual and spatial tasks. [TODO: Check that each task maps to one primary category.]

Environment. Tasks run in Ubuntu 22.04 virtual machines at a screen resolution of 1920×1080, with Chrome, LibreOffice, a file manager and locally hosted web apps. Each task ships with a setup script that restores a snapshot and a checker that inspects the final application or file state. [TODO: Environment details, application versions, and whether we build on an existing harness such as OSWorld.]

Category	Example task	Tasks	Human time (s)	Tolerance τ (s)
Continuous manipulation	Drag the invoice file into the *2026* folder; set the brightness slider to 40%	XX	XX	XX
Scrolling and hidden content	Open the last entry in the sidebar's nested list	XX	XX	XX
Transient UI	Choose *Duplicate* from the hover menu of the third row	XX	XX	XX
Custom widgets	Set the start date in the date picker to the first Monday of next month	XX	XX	XX
Text selection and editing	Make only the second sentence of the paragraph bold	XX	XX	XX
Interruptions	Close the newsletter pop-up, then add the item to the cart	XX	XX	XX
Keyboard and windows	Copy the order number from the e-mail into the refund form	XX	XX	XX
Visual and spatial	Rotate the shape on the canvas so its arrow points down	XX	XX	XX
Total		**75**	XX	XX

Table 1. Task categories in LastStraw with an example task, the number of tasks, and the median human completion time and median tolerance across annotators. [TODO: Confirm the category scheme and examples against the 75 tasks.]

Selection. [TODO: State whether tasks were filtered against any model during authoring (for example, keeping only tasks on which a pilot agent struggled). If so, name the pilot model and report its results separately to avoid inflating its frustration rate.]

3.3 Human effort and tolerance labels

Effort. Each task was performed by n annotators who did not write it. We record the time from the first view of the task to completion and the number of input actions (clicks, drags, scrolls, key presses and typed strings). For task i, the human effort is the median completion time t_i^H and the median action count a_i^H across annotators. Annotators completed XX% of attempts on the first try.

Tolerance. After completing a task, annotators answered two questions: *"If you had asked an assistant to do this task while you watched the screen, how long would you wait before stopping it and doing it yourself?"* and *"How many actions that do not move toward the goal would you accept before stopping it?"* The task's time tolerance τ_i and action tolerance κ_i are the medians of these answers. Across tasks, the median τ_i is XX seconds, or X.X$\times$ the human completion time. [TODO: Decide how wasted actions are counted against κ_i, or switch κ_i to a total-action budget.]

Validating stated tolerance. Stated tolerance can differ from behaviour. On XX tasks, n annotators watched replays of agent runs and pressed a button at the moment they would take over. [TODO: Report how behavioural take-over times compare with the stated τ_i, and inter-annotator agreement on τ_i.]

3.4 Metrics

For a run r on task i, let $c_r \in \{0,1\}$ be the checker's verdict on the final state, T_r the wall-clock time from the instruction to the agent's final action or completion message, and A_r the number of actions it took. Runs are capped at XX actions or XX minutes, well above every task's budget, so that we can also measure correctness without a budget.

Success rate (SR). The fraction of runs with $c_r = 1$, as reported by existing benchmarks.

Tolerated success rate (TSR). A run succeeds only if it is correct and within both budgets:

$$s_r = c_r \cdot \mathbb{1}[T_r \le \tau_i] \cdot \mathbb{1}[A_r \le \kappa_i], \qquad TSR = \frac{1}{|R|} \sum_{r \in R} s_r. \tag{1}$$

The *frustration rate* is $FR = 1 - TSR$. Every run that fails, stops early, or runs past the point at which the user would have taken over counts as a frustration.

Tolerance gap. $\Delta = SR - TSR$ is the share of runs that a correctness-only benchmark counts as successes but a user would not.

Effort ratios. On correct runs we report the median time ratio T_r / t_i^H and action ratio A_r / a_i^H, which show how far agents are from human effort even when they succeed.

Consistency. A user who is frustrated once may stop delegating the task. Following pass^k [43], TSR^k is the fraction of tasks on which all k runs are tolerated successes.

3.5 Frustration patterns

The metrics say whether a run frustrates, and the patterns in Table 2 say how. Looping, stalling and detours are detected from the action log and screenshots with fixed rules; primitive failures, grounding misses and unneeded hand-backs are labelled by an LLM judge whose labels annotators verify on XX runs; false completions and collateral changes come from the checker and a state diff. [TODO: Report detector precision and recall against human labels (Appendix C).]

4 Experiments

4.1 Setup

Models. We evaluate M frontier models with computer-use support: Claude Fable 5.1, Claude Opus 5.5, Claude Sonnet 5.5, GPT-6-Astra, Gemini 3.8 and an open-weight CUA. [TODO: Final model list, versions, reasoning effort and access dates.]

Harness. Closed models run with their providers' native computer-use tools; open-weight models run in a shared screenshot-and-action harness. The agent receives the task text and a screenshot after every action. We run each model k times per task, on identical virtual machines in the same region,

Pattern	What the user sees	Detector
Looping	The agent repeats an action, or a short cycle of actions, without the screen changing in a way that brings it closer to the goal.	Repeated action n-grams on near-identical screenshots
Stalling	Long pauses before trivial actions, so the screen stays still while the agent deliberates.	Inter-action gap $> X$ s
Detour	The agent reaches the goal by a much longer path than needed, e.g. menus and search instead of a visible control.	$A > X \times a^H$ on correct runs
Primitive failure	The agent cannot perform a basic interaction such as a drag, a hover or a scroll inside a panel, and retries it.	LLM judge, human-verified
Grounding miss	Clicks or types into the wrong element, sometimes repeatedly.	LLM judge, human-verified
False completion	The agent reports success although the end state is wrong.	Checker fails after a "done" claim
Unneeded hand-back	The agent stops to ask the user for confirmation or input that the task does not require, or gives up.	LLM judge, human-verified
Collateral change	The agent completes the task but changes something else that the user must undo.	State diff outside the task scope

Table 2. Frustration patterns. A run can show several patterns; patterns explain why a run fails, while the pass criterion itself depends only on correctness and the tolerance budget (§3.4). [TODO: Finalize detectors and thresholds; report agreement with human labels in Appendix C.]

and log screenshots, actions and timestamps. [TODO: Decide whether all models also run in the shared harness, to separate model from harness effects.]

Human reference. Human SR and TSR come from the annotators' effort sessions (§3.3), scored with the same checker and budgets.

4.2 Correctness overstates usability

Table 3 reports both metrics. The strongest model reaches XX.X% SR but only XX.X% TSR, so XX points of its apparent success come from runs that a user would have stopped. Human annotators reach XX.X% TSR. [TODO: Two to three sentences on the main pattern once results are in.]

Rankings change. Ordering models by TSR instead of SR changes the ranking (Kendall's $\tau_b = X.XX$). [TODO: Name the largest rank changes and explain them, e.g. a slower but more accurate model dropping below a faster one.]

4.3 How far agents are from human effort

Even on correct runs, agents take a median X.X× the human time and X.X× the human actions (Table 3, Figure 4). [TODO: Split time into model latency and environment time; report whether higher reasoning effort raises SR but lowers TSR.]

Model	Harness	SR (%)	TSR (%)	Gap (pts)	Time ratio	Action ratio
Claude Fable 5.1	native CU tool	XX.X	XX.X	XX.X	X.X×	X.X×
Claude Opus 5.5	native CU tool	XX.X	XX.X	XX.X	X.X×	X.X×
Claude Sonnet 5.5	native CU tool	XX.X	XX.X	XX.X	X.X×	X.X×
GPT-6-Astra	native CU tool	XX.X	XX.X	XX.X	X.X×	X.X×
Gemini 3.8	native CU tool	XX.X	XX.X	XX.X	X.X×	X.X×
Open-weight CUA	shared harness	XX.X	XX.X	XX.X	X.X×	X.X×
Human annotators	–	XX.X	XX.X	XX.X	1.0×	1.0×

Table 3. Main results on the 75 tasks, averaged over k runs per task. SR counts a run as successful if the end state is correct, whatever it took; TSR additionally requires the run to stay within the human time and action budgets. Gap is $SR - TSR$: the share of runs that are correct but beyond human tolerance. Time and action ratios are medians of agent over human effort on correct runs. [TODO: Final model list, reasoning effort settings and 95% CIs.]

Figure placeholder: SR (x) vs. TSR (y) per model, with the diagonal; distance below the diagonal is the tolerance gap.

Figure 3. Correctness versus tolerated success. Every model lies below the diagonal; the vertical distance is the share of correct runs that exceed human tolerance. [TODO: Make figure.]

4.4 Where frustration comes from

Table 4 breaks down frustrated runs by pattern. Looping is the most common pattern, occurring in XX% of frustrated runs. [TODO: Per-category breakdown (Table 1 categories × models), the most and least frustrating categories, and whether patterns are shared across model families.]

4.5 Case studies

[TODO: Two or three short trajectories, each with a screenshot strip: (1) a loop on a hover menu, (2) a correct but intolerably slow run on a date picker, (3) a false completion after a pop-up.]

5 Discussion

Correct is not the same as useful. [TODO: What the tolerance gap means for how CUA progress is reported, and why TSR should be reported alongside SR.]

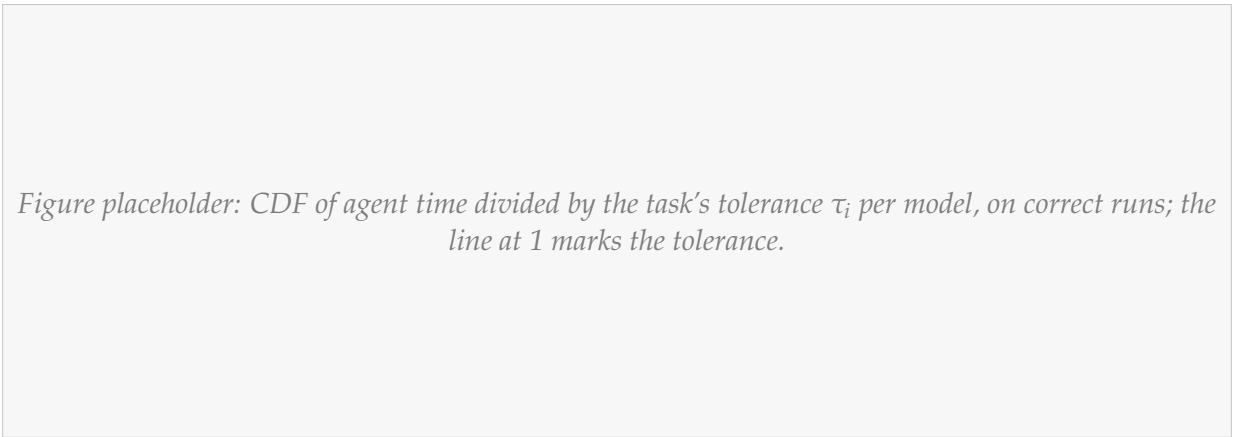


Figure 4. Agent time relative to human tolerance on correct runs. [TODO: Make figure.]

Model	Looping	Stalling	Detour	Primitive	Grounding	False done	Hand-back	Collateral
Claude Fable 5.1	XX	XX	XX	XX	XX	XX	XX	XX
Claude Opus 5.5	XX	XX	XX	XX	XX	XX	XX	XX
Claude Sonnet 5.5	XX	XX	XX	XX	XX	XX	XX	XX
GPT-6-Astra	XX	XX	XX	XX	XX	XX	XX	XX
Gemini 3.8	XX	XX	XX	XX	XX	XX	XX	XX
Open-weight CUA	XX	XX	XX	XX	XX	XX	XX	XX

Table 4. Percent of runs in which each frustration pattern (Table 2) occurs. A run can show several patterns.

Tolerance as a training signal. [TODO: Whether time and action budgets can serve as a reward or a stopping criterion during training, and the risk that agents trade correctness for speed.]

When to hand back control. [TODO: An agent that recognizes a loop early and hands the task back may frustrate less than one that eventually succeeds. Discuss whether a timely hand-back should earn partial credit.]

6 Conclusion

We introduced LastStraw, a benchmark of 75 human-written computer-use tasks that are easy for people, each labelled with the effort it takes a person and the patience a person would extend to an agent. Scoring a run as successful only when it is correct and within that tolerance shows that frontier agents frustrate users on XX% of tasks that people finish in seconds, a gap that correctness-only benchmarks do not show. [TODO: Final sentence on release and future work.]

Limitations

Wall-clock time depends on infrastructure. Agent time includes model latency, which varies with provider load, region and hardware. We fix the infrastructure across runs and also report action-based budgets, which do not depend on latency. [TODO: Report the variance of T_r across repeated runs.]

Tolerance is subjective. Tolerance varies across people, tasks and contexts: a user watching the screen tolerates less than one who delegates a task and returns later. Our labels describe the attended setting with n annotators per task.

Scale. 75 tasks support the comparisons we make but not fine-grained claims about individual categories. [TODO: Report confidence intervals per category.]

Ethics Statement

Annotators were contracted and paid at or above local minimum wage; no personal data about them is recorded or released. Tasks run in isolated virtual machines and locally hosted applications, and no task acts on live third-party services or real user accounts.

References

[1] Anthropic. Developing a computer use model. https://www.anthropic.com/news/developing-computer-use, 2024. URL https://www.anthropic.com/news/developing-computer-use. Blog post, October 22, 2024.

[2] OpenAI. Computer-using agent. https://openai.com/index/computer-using-agent/, 2025. URL https://openai.com/index/computer-using-agent/. Blog post, January 23, 2025 (the model behind Operator).

[3] Google DeepMind. Introducing the Gemini 2.5 Computer Use model. https://blog.google/technology/google-deepmind/gemini-computer-use-model/, 2025. URL https://blog.google/technology/google-deepmind/gemini-computer-use-model/. Blog post, October 7, 2025.

[4] Yujia Qin, Yining Ye, Junjie Fang, Haoming Wang, Shihao Liang, Shizuo Tian, Junda Zhang, Jiahao Li, Yunxin Li, Shijue Huang, Wanjun Zhong, Kuanye Li, et al. UI-TARS: Pioneering Automated GUI Interaction with Native Agents. *arXiv preprint arXiv:2501.12326*, 2025. URL https://arxiv.org/abs/2501.12326.

[5] Haoming Wang, Haoyang Zou, Huatong Song, Jiazhan Feng, Junjie Fang, Junting Lu, Longxiang Liu, Qinyu Luo, Shihao Liang, Shijue Huang, Wanjun Zhong, Yining Ye, et al. UI-TARS-2 Technical Report: Advancing GUI Agent with Multi-Turn Reinforcement Learning. *arXiv preprint arXiv:2509.02544*, 2025. URL https://arxiv.org/abs/2509.02544.

[6] Xinyuan Wang, Bowen Wang, Dunjie Lu, Junlin Yang, Tianbao Xie, Junli Wang, Jiaqi Deng, Xiaole Guo, Yiheng Xu, Chen Henry Wu, Zhennan Shen, Zhuokai Li, et al. OpenCUA: Open Foundations for Computer-Use Agents. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. URL https://arxiv.org/abs/2508.09123.

[7] Tianbao Xie, Danyang Zhang, Jixuan Chen, Xiaochuan Li, Siheng Zhao, Ruisheng Cao, Toh Jing Hua, Zhoujun Cheng, Dongchan Shin, Fangyu Lei, Yitao Liu, Yiheng Xu, et al. OSWorld: Benchmarking Multimodal Agents for Open-Ended Tasks in Real Computer Environments. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2024. URL https://arxiv.org/abs/2404.07972.

[8] Pranjal Aggarwal, Lawrence Keunho Jang, Sean Welleck, Daniel Fried, Ruslan Salakhutdinov, and Jing Yu Koh. cua-speedrun: Standardized Benchmarking of the Speed of Computer-Use Agents. *arXiv preprint arXiv:2609.40284*, 2026. URL https://arxiv.org/abs/2609.40284.

[9] Shuyan Zhou, Frank F. Xu, Hao Zhu, Xuhui Zhou, Robert Lo, Abishek Sridhar, Xianyi Cheng, Tianyue Ou, Yonatan Bisk, Daniel Fried, Uri Alon, and Graham Neubig. WebArena: A Realistic Web Environment for Building Autonomous Agents. In *International Conference on Learning Representations (ICLR)*, 2024. URL https://arxiv.org/abs/2307.13854.

[10] Christopher Rawles, Sarah Clinckemaillie, Yifan Chang, Jonathan Waltz, Gabrielle Lau, Marybeth Fair, Alice Li, William Bishop, Wei Li, Folawiyo Campbell-Ajala, Daniel Toyama, Robert Berry, et al. AndroidWorld: A Dynamic Benchmarking Environment for Autonomous Agents. In *International Conference on Learning Representations (ICLR)*, 2025. URL https://arxiv.org/abs/2405.14573.

[11] Tianci Xue, Weijian Qi, Tianneng Shi, Chan Hee Song, Boyu Gou, Dawn Song, Huan Sun, and Yu Su. An Illusion of Progress? Assessing the Current State of Web Agents. In *Conference on Language Modeling (COLM)*, 2025. URL https://arxiv.org/abs/2504.01382.

[12] Reyna Abhyankar, Qi Qi, and Yiying Zhang. OSWorld-Human: Benchmarking the Efficiency of Computer-Use Agents. In *Proceedings of Machine Learning and Systems (MLSys)*, 2026. URL https://arxiv.org/abs/2506.16042.

[13] Tianchen Guan, Xinlei Lin, Royce Cheng-Yue, Xiangjun Wang, and Shuyan Zhou. Component-Bench: Diagnosing Component-Level Failures in Computer-Use Agents. In *Conference on Language Modeling (COLM)*, 2026. URL https://arxiv.org/abs/2608.18307.

[14] Irina Ceaparu, Jonathan Lazar, Katie Bessiere, John Robinson, and Ben Shneiderman. Determining causes and severity of end-user frustration. *International Journal of Human-Computer Interaction*, 17 (3):333–356, 2004. doi: 10.1207/s15327590ijhc1703_3.

[15] Jonathan Lazar, Adam Jones, Mary Hackley, and Ben Shneiderman. Severity and impact of computer user frustration: A comparison of student and workplace users. *Interacting with Computers*, 18(2):187–207, 2006. doi: 10.1016/j.intcom.2005.06.001.

[16] Robert B. Miller. Response time in man-computer conversational transactions. In *Proceedings of the December 9–11, 1968, Fall Joint Computer Conference, Part I (AFIPS '68)*, volume 33, pages 267–277. ACM, 1968. doi: 10.1145/1476589.1476628.

[17] Jakob Nielsen. *Usability Engineering*. Academic Press, 1993. URL https://www.nngroup.com/articles/response-times-3-important-limits/. Chapter 5, response-time limits.

[18] Raja Parasuraman and Victor Riley. Humans and automation: Use, misuse, disuse, abuse. *Human Factors*, 39(2):230–253, 1997. doi: 10.1518/001872097778543886.

[19] Berkeley J. Dietvorst, Joseph P. Simmons, and Cade Massey. Algorithm aversion: People erroneously avoid algorithms after seeing them err. *Journal of Experimental Psychology: General*, 144(1): 114–126, 2015. doi: 10.1037/xge0000033.

[20] Hussein Mozannar, Gagan Bansal, Cheng Tan, Adam Fourney, Victor Dibia, Jingya Chen, Jack Gerrits, Tyler Payne, Matheus Kunzler Maldaner, Madeleine Grunde-McLaughlin, Eric Zhu, Griffin Bassman, Jacob Alber, Peter Chang, Ricky Loynd, Friederike Niedtner, Ece Kamar, Maya Murad,

Rafah Hosn, and Saleema Amershi. Magentic-UI: Towards human-in-the-loop agentic systems. *arXiv preprint arXiv:2507.22358*, 2025. URL https://arxiv.org/abs/2507.22358. Also Microsoft Research Technical Report MSR-TR-2025-40.

[21] Pradyumna Shome, Sashreek Krishnan, and Sauvik Das. Why Johnny can't use agents: Industry aspirations vs. user realities with AI agents. In *Proceedings of the ACM Conference on AI and Agentic Systems (CAIS '26)*, pages 597–619. ACM, 2026. doi: 10.1145/3786335.3813140. arXiv:2509.14528.

[22] Kaixin Li, Ziyang Meng, Hongzhan Lin, Ziyang Luo, Yuchen Tian, Jing Ma, Zhiyong Huang, and Tat-Seng Chua. ScreenSpot-Pro: GUI Grounding for Professional High-Resolution Computer Use. In *Proceedings of the 33rd ACM International Conference on Multimedia (MM)*, 2025. URL https://arxiv.org/abs/2504.07981.

[23] Yanzhe Zhang, Tao Yu, and Diyi Yang. Attacking Vision-Language Computer Agents via Pop-ups. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2025. URL https://arxiv.org/abs/2411.02391.

[24] Hans Moravec. *Mind Children: The Future of Robot and Human Intelligence*. Harvard University Press, Cambridge, MA, 1988. ISBN 0674576160.

[25] Frank F. Xu, Yufan Song, Boxuan Li, Yuxuan Tang, Kritanjali Jain, Mengxue Bao, Zora Z. Wang, Xuhui Zhou, Zhitong Guo, Murong Cao, Mingyang Yang, Hao Yang Lu, et al. TheAgentCompany: Benchmarking LLM Agents on Consequential Real World Tasks. *arXiv preprint arXiv:2412.14161*, 2024. URL https://arxiv.org/abs/2412.14161.

[26] Mengqi Yuan, Zilong Zhou, Xinzhuang Xiong, Weiming Wu, Jiayang Sun, Jiamin Song, Kaiqian Cui, Bowen Wang, Haoyuan Wu, Yitong Li, Dunjie Lu, Haikong Lu, et al. OSWorld 2.0: Benchmarking Computer Use Agents on Long-Horizon Real-World Tasks. *arXiv preprint arXiv:2606.29537*, 2026. URL https://arxiv.org/abs/2606.29537.

[27] ARC Prize Foundation. ARC-AGI-3: A New Challenge for Frontier Agentic Intelligence. *arXiv preprint arXiv:2603.24621*, 2026. URL https://arxiv.org/abs/2603.24621.

[28] Rogerio Bonatti, Dan Zhao, Francesco Bonacci, Dillon Dupont, Sara Abdali, Yinheng Li, Yadong Lu, Justin Wagle, Kazuhito Koishida, Arthur Bucker, Lawrence Jang, and Zack Hui. Windows Agent Arena: Evaluating Multi-Modal OS Agents at Scale. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, 2025. URL https://arxiv.org/abs/2409.08264.

[29] Pei Yang, Hai Ci, and Mike Zheng Shou. macOSWorld: A Multilingual Interactive Benchmark for GUI Agents. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. URL https://arxiv.org/abs/2506.04135.

[30] Mariya Davydova, Daniel Jeffries, Patrick Barker, Arturo Márquez Flores, and Sinéad Ryan. OS-Universe: Benchmark for Multimodal GUI-navigation AI Agents. *arXiv preprint arXiv:2505.03570*, 2025. URL https://arxiv.org/abs/2505.03570.

[31] Jing Yu Koh, Robert Lo, Lawrence Jang, Vikram Duvvur, Ming Chong Lim, Po-Yu Huang, Graham Neubig, Shuyan Zhou, Ruslan Salakhutdinov, and Daniel Fried. VisualWebArena: Evaluating Multimodal Agents on Realistic Visual Web Tasks. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024. URL https://arxiv.org/abs/2401.13649.

[32] Hongliang He, Wenlin Yao, Kaixin Ma, Wenhao Yu, Yong Dai, Hongming Zhang, Zhenzhong Lan, and Dong Yu. WebVoyager: Building an End-to-End Web Agent with Large Multimodal Models. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*, 2024. URL https://arxiv.org/abs/2401.13919.

[33] Alexandre Drouin, Maxime Gasse, Massimo Caccia, Issam H. Laradji, Manuel Del Verme, Tom Marty, Léo Boisvert, Megh Thakkar, Quentin Cappart, David Vazquez, Nicolas Chapados, and Alexandre Lacoste. WorkArena: How Capable Are Web Agents at Solving Common Knowledge Work Tasks? In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, 2024. URL https://arxiv.org/abs/2403.07718.

[34] Léo Boisvert, Megh Thakkar, Maxime Gasse, Massimo Caccia, Thibault Le Sellier De Chezelles, Quentin Cappart, Nicolas Chapados, Alexandre Lacoste, and Alexandre Drouin. WorkArena++: Towards Compositional Planning and Reasoning-based Common Knowledge Work Tasks. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2024. URL https://arxiv.org/abs/2407.05291.

[35] Xiang Deng, Yu Gu, Boyuan Zheng, Shijie Chen, Samuel Stevens, Boshi Wang, Huan Sun, and Yu Su. Mind2Web: Towards a Generalist Agent for the Web. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2023. URL https://arxiv.org/abs/2306.06070.

[36] Xing Han Lù, Amirhossein Kazemnejad, Nicholas Meade, Arkil Patel, Dongchan Shin, Alejandra Zambrano, Karolina Staczak, Peter Shaw, Christopher J. Pal, and Siva Reddy. AgentRewardBench: Evaluating Automatic Evaluations of Web Agent Trajectories. *arXiv preprint arXiv:2504.08942*, 2025. URL https://arxiv.org/abs/2504.08942.

[37] XLANG Lab. Introducing OSWorld-Verified. https://xlang.ai/blog/osworld-verified, 2025. URL https://xlang.ai/blog/osworld-verified. Blog post, July 28, 2025.

[38] Saaket Agashe, Kyle Wong, Vincent Tu, Jiachen Yang, Ang Li, and Xin Eric Wang. Agent S2: A Compositional Generalist-Specialist Framework for Computer Use Agents. *arXiv preprint arXiv:2504.00906*, 2025. URL https://arxiv.org/abs/2504.00906.

[39] Jingqi Yang, Zhilong Song, Jiawei Chen, Mingli Song, Sheng Zhou, linjun sun, Xiaogang Ouyang, Chun Chen, and Can Wang. GUI-Robust: A Comprehensive Dataset for Testing GUI Agent Robustness in Real-World Anomalies. In *Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2026. URL https://arxiv.org/abs/2506.14477.

[40] Haoming Li, Defu Cao, Xu Hu, Yingtao Luo, Yiyang Wang, Xianghe Xu, Jiashu He, Wanting Jiang, Yifan Zhang, Yunhan Zhao, Siyang Wu, Chen Yang, et al. PersonaCUABench: Benchmarking Computer-Use Agents Serving Diverse Personas. *arXiv preprint arXiv:2610.22990*, 2026. URL https://arxiv.org/abs/2610.22990.

[41] Sayash Kapoor, Benedikt Stroebl, Zachary S. Siegel, Nitya Nadgir, and Arvind Narayanan. AI Agents That Matter. *Transactions on Machine Learning Research (TMLR)*, 2025. URL https://arxiv.org/abs/2407.01502.

[42] Sayash Kapoor, Benedikt Stroebl, Peter Kirgis, Nitya Nadgir, Zachary S Siegel, Boyi Wei, Tianci Xue, Ziru Chen, Felix Chen, Saiteja Utpala, Franck Ndzomga, Dheeraj Oruganty, et al. Holistic

Agent Leaderboard: The Missing Infrastructure for AI Agent Evaluation. In *International Conference on Learning Representations (ICLR)*, 2026. URL https://arxiv.org/abs/2510.11977.

[43] Shunyu Yao, Noah Shinn, Pedram Razavi, and Karthik Narasimhan. τ-bench: A Benchmark for Tool-Agent-User Interaction in Real-World Domains. In *International Conference on Learning Representations (ICLR)*, 2025. URL https://arxiv.org/abs/2406.12045.

[44] Stephan Rabanser, Sayash Kapoor, Peter Kirgis, Kangheng Liu, Saiteja Utpala, and Arvind Narayanan. Towards a Science of AI Agent Reliability. In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, 2026. URL https://arxiv.org/abs/2602.16666.

[45] Yuxuan Zhu, Tengjun Jin, Yada Pruksachatkun, Andy Zhang, Shu Liu, Sasha Cui, Sayash Kapoor, Shayne Longpre, Kevin Meng, Rebecca Weiss, Fazl Barez, Rahul Gupta, et al. Establishing Best Practices in Building Rigorous Agentic Benchmarks. In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2025. URL https://arxiv.org/abs/2507.02825.

[46] Thomas Kwa, Ben West, Joel Becker, Amy Deng, Katharyn Garcia, Max Hasin, Sami Jawhar, Megan Kinniment, Nate Rush, Sydney Von Arx, Ryan Bloom, Thomas Broadley, et al. Measuring AI Ability to Complete Long Software Tasks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2025. URL https://arxiv.org/abs/2503.14499. arXiv v1 title: Measuring AI Ability to Complete Long Tasks.

[47] George Thomas, Alex J. Chan, Jikun Kang, Wenqi Wu, Filippos Christianos, Fraser Greenlee, Andy Toulis, and Marvin Purtorab. WebGames: Challenging General-Purpose Web-Browsing AI Agents. *arXiv preprint arXiv:2502.18356*, 2025. URL https://arxiv.org/abs/2502.18356.

[48] Yixiao Song, Katherine Thai, Chau Minh Pham, Yapei Chang, Mazin Nadaf, and Mohit Iyyer. BEARCUBS: A benchmark for computer-using web agents. *arXiv preprint arXiv:2503.07919*, 2025. URL https://arxiv.org/abs/2503.07919.

[49] François Chollet. On the Measure of Intelligence. *arXiv preprint arXiv:1911.01547*, 2019. URL https://arxiv.org/abs/1911.01547.

[50] Francois Chollet, Mike Knoop, Gregory Kamradt, Bryan Landers, and Henry Pinkard. ARC-AGI-2: A New Challenge for Frontier AI Reasoning Systems. *arXiv preprint arXiv:2505.11831*, 2025. URL https://arxiv.org/abs/2505.11831.

[51] SimpleBench Team. SimpleBench: Everyday reasoning benchmark where non-specialized humans outperform frontier models. https://simple-bench.com/, 2024. URL https://simple-bench.com/. Benchmark website and leaderboard; public release October 2024.

[52] Mert Cemri, Melissa Z. Pan, Shuyi Yang, Lakshya A. Agrawal, Bhavya Chopra, Rishabh Tiwari, Kurt Keutzer, Aditya Parameswaran, Dan Klein, Kannan Ramchandran, Matei Zaharia, Joseph E. Gonzalez, et al. Why Do Multi-Agent LLM Systems Fail? In *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, 2025. URL https://arxiv.org/abs/2503.13657.

[53] Kunlun Zhu, Zijia Liu, Bingxuan Li, Muxin Tian, Yingxuan Yang, Jiaxun Zhang, Pengrui Han, Qipeng Xie, Fuyang Cui, Weijia Zhang, Xiaoteng Ma, Xiaodong Yu, et al. Where LLM Agents Fail and How They can Learn From Failures. *arXiv preprint arXiv:2509.25370*, 2025. URL https://arxiv.org/abs/2509.25370.

[54] Xingyu Chen, Jiahao Xu, Tian Liang, Zhiwei He, Jianhui Pang, Dian Yu, Linfeng Song, Qiuzhi Liu, Mengfei Zhou, Zhuosheng Zhang, Rui Wang, Zhaopeng Tu, et al. Do NOT Think That Much for 2+3=? On the Overthinking of Long Reasoning Models. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, volume 267 of *PMLR*, pages 9487–9499, 2025. URL https://arxiv.org/abs/2412.21187.

[55] Alejandro Cuadron, Dacheng Li, Wenjie Ma, Xingyao Wang, Yichuan Wang, Siyuan Zhuang, Shu Liu, Luis Gaspar Schroeder, Tian Xia, Huanzhi Mao, Nicholas Thumiger, Aditya Desai, et al. The Danger of Overthinking: Examining the Reasoning-Action Dilemma in Agentic Tasks. *arXiv preprint arXiv:2502.08235*, 2025. URL https://arxiv.org/abs/2502.08235.

[56] Qijun Han, Haoqin Tu, Zijun Wang, Haoyue Dai, Yiyang Zhou, Nancy Lau, Alvaro A. Cardenas, Yuhui Xu, Ran Xu, Caiming Xiong, Zeyu Zheng, Huaxiu Yao, et al. VLAA-GUI: Knowing When to Stop, Recover, and Search, A Modular Framework for GUI Automation. *arXiv preprint arXiv:2604.21375*, 2026. URL https://arxiv.org/abs/2604.21375.

[57] Jinbiao Wei, Kangqi Ni, Yilun Zhao, Guo Gan, and Arman Cohan. Step-level Optimization for Efficient Computer-use Agents. *arXiv preprint arXiv:2604.27151*, 2026. URL https://arxiv.org/abs/2604.27151.

[58] Katie Bessière, John E. Newhagen, John P. Robinson, and Ben Shneiderman. A model for computer frustration: The role of instrumental and dispositional factors on incident, session, and post-session frustration and mood. *Computers in Human Behavior*, 22(6):941–961, 2006. doi: 10.1016/j.chb.2004.03.015.

[59] Stuart K. Card, George G. Robertson, and Jock D. Mackinlay. The information visualizer, an information workspace. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '91)*, pages 181–186. ACM, 1991. doi: 10.1145/108844.108874.

[60] Fiona Fui-Hoon Nah. A study on tolerable waiting time: How long are Web users willing to wait? *Behaviour & Information Technology*, 23(3):153–163, 2004. doi: 10.1080/01449290410001669914. URL https://doi.org/10.1080/01449290410001669914.

[61] Dennis F. Galletta, Raymond Henry, Scott McCoy, and Peter Polak. Web site delays: How tolerant are users? *Journal of the Association for Information Systems*, 5(1):1–28, 2004. doi: 10.17705/1jais.00044.

[62] Ben Shneiderman. Response time and display rate in human performance with computers. *ACM Computing Surveys*, 16(3):265–285, 1984. doi: 10.1145/2514.2517.

[63] Jim Dabrowski and Ethan V. Munson. 40 years of searching for the best computer system response time. *Interacting with Computers*, 23(5):555–564, 2011. doi: 10.1016/j.intcom.2011.05.008.

[64] Stuart K. Card, Thomas P. Moran, and Allen Newell. The keystroke-level model for user performance time with interactive systems. *Communications of the ACM*, 23(7):396–410, 1980. doi: 10.1145/358886.358895.

[65] Stuart K. Card, Thomas P. Moran, and Allen Newell. *The Psychology of Human-Computer Interaction*. Lawrence Erlbaum Associates, Hillsdale, NJ, 1983.

[66] John D. Lee and Katrina A. See. Trust in automation: Designing for appropriate reliance. *Human Factors*, 46(1):50–80, 2004. doi: 10.1518/hfes.46.1.50_30392.

[67] Eric Horvitz. Principles of mixed-initiative user interfaces. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '99)*, pages 159–166. ACM, 1999. doi: 10.1145/302979.303030.

[68] Saleema Amershi, Dan Weld, Mihaela Vorvoreanu, Adam Fourney, Besmira Nushi, Penny Collisson, Jina Suh, Shamsi Iqbal, Paul N. Bennett, Kori Inkpen, Jaime Teevan, Ruth Kikin-Gil, and Eric Horvitz. Guidelines for human-AI interaction. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (CHI '19)*, pages 1–13. ACM, 2019. doi: 10.1145/3290605.3300233.

[69] Mykola Maslych, Mohammadreza Katebi, Christopher Lee, Yahya Hmaiti, Amirpouya Ghasemaghaei, Christian Pumarada, Janneese Palmer, Esteban Segarra Martinez, Marco Emporio, Warren Snipes, Ryan P. McMahan, and Jr. LaViola, Joseph J. Mitigating response delays in free-form conversations with LLM-powered intelligent virtual agents. In *Proceedings of the 7th ACM Conference on Conversational User Interfaces (CUI '25)*, pages 1–15. ACM, 2025. doi: 10.1145/3719160.3736636.

[70] Shuning Zhang, Jingruo Chen, Zhiqi Gao, Jiajing Gao, Xin Yi, and Hewu Li. Characterizing unintended consequences in human-GUI agent collaboration for web browsing. *arXiv preprint arXiv:2505.09875*, 2025. URL https://arxiv.org/abs/2505.09875.

[71] Faria Huq, Zora Zhiruo Wang, Frank F. Xu, Tianyue Ou, Shuyan Zhou, Jeffrey P. Bigham, and Graham Neubig. CowPilot: A framework for autonomous and human-agent collaborative web navigation. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (System Demonstrations)*, pages 163–172. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.naacl-demo.17.

[72] Faria Huq, Zora Zhiruo Wang, Zhanqiu Guo, Venu Arvind Arangarajan, Tianyue Ou, Frank F. Xu, Shuyan Zhou, Graham Neubig, and Jeffrey P. Bigham. Modeling distinct human interaction in web agents. In *Findings of the Association for Computational Linguistics: EMNLP 2026*, pages 43195–43212. Association for Computational Linguistics, 2026. doi: 10.18653/2026.findings-emnlp.2029. URL https://aclanthology.org/2026.findings-emnlp.2029. arXiv:2602.17588.

[73] Chaoran Chen, Zhiping Zhang, Zeya Chen, Eryue Xu, Yinuo Yang, Ibrahim Khalilov, Simret A Gebreegziabher, Yanfang Ye, Ziang Xiao, Yaxing Yao, Tianshi Li, and Toby Jia-Jun Li. Comparing Human Oversight Strategies for Computer-Use Agents. *arXiv preprint arXiv:2604.04918*, 2026. URL https://arxiv.org/abs/2604.04918.

[74] Felicia Fang-Yi Tan, Moritz Alexander Messerschmidt, Wen Yin, and Oded Nov. The impact of response latency and task type on human-LLM interaction and perception. In *Proceedings of the 2026 CHI Conference on Human Factors in Computing Systems (CHI '26)*, pages 1–17. ACM, 2026. doi: 10.1145/3772318.3790716. arXiv:2604.06183.

A Task Authoring Guidelines

[TODO: Guidelines given to task authors, review rounds, and rejected task types.]

B Annotation Protocol

[TODO: Effort sessions (recording, action logging), tolerance questions verbatim, the replay take-over study, annotator pool and pay, and inter-annotator agreement.]

C Frustration Pattern Detectors

[TODO: Rules and thresholds for looping, stalling and detours; the LLM-judge prompt for primitive failures, grounding misses and hand-backs; precision and recall against human labels.]

D Full Task List

[TODO: All 75 tasks with category, human time t^H, actions a^H, tolerance τ and κ, and per-model TSR.]

E Harness and Model Settings

[TODO: Provider tools, prompts, screen resolution, action space, run caps and reasoning effort for each model.]